

Schizophrenia Recognition from Eye Movements: A Hierarchical Multi-Tier Machine Learning Framework

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Abstract—Timely and accurate diagnosis of Schizophrenia (SZ) remains a major challenge in psychiatry due to the subjective nature of qualitative clinical questionnaires. Free-viewing eye-tracking paradigms offer a non-invasive, cost-effective, and quantitative alternative by capturing cognitive and visual attention abnormalities. In this study, we propose a Hierarchical 5-Tier Machine Learning Framework for eye movement-based SZ recognition using the large-scale EMS dataset (208 subjects). Our pipeline integrates: spatiotemporal cleaning and pupil diameter preservation (Tier 1); stimulus-level and contextual delta feature engineering (Tier 2); tabular ensemble baselines (Tier 3); deep sequential models including Bidirectional Cross-Attention Hybrid Stream (BiCA-HS) and Graph Attention Network with Cross-Attention Enhanced Fusion (GNN-CEFAM) (Tier 4); and a meta-learner stacking classifier (Tier 5). To prevent visual representation collapse in deep models, we resolve mathematical degeneracies in cross-attention formulations. Evaluation under strict, no-leakage GroupKFold cross-validation demonstrates that our Tier 5 Meta-Learner achieves an out-of-fold Area Under the Curve (AUC) of 0.9628 ± 0.0024 and Accuracy of $87.50\% \pm 1.02\%$, while the deep sequential BiCA-HS reaches an AUC of 0.9454 ± 0.0099 and a high Sensitivity of $96.25\% \pm 1.77\%$. Both models significantly outperform the current State-of-the-Art MSNet (AUC of 0.8854 and Accuracy of 81.25%), validated by statistical bootstrap significance tests ($p < 0.01$). Explainability analyses using SHAP and cross-attention saliency mapping demonstrate clinical alignment, showing that schizophrenics display restricted visual exploration and background attentional dispersion.

Index Terms—Schizophrenia recognition, Eye movements, Spatiotemporal GNN, Cross-attention fusion, Tabular baseline, Meta-learner, Explainable AI.

I. INTRODUCTION

Schizophrenia (SZ) is a severe, chronic mental disorder characterized by cognitive deficits, flat affect, delusions, and compromised social functioning. According to psychiatric statistics, SZ is a major contributor to global disability, imposing heavy psychological and economic burdens on families and society. Early screening and clinical intervention are critical to improving patient outcomes. Currently, clinical diagnosis relies heavily on symptomatology assessments (e.g., DSM-V or

ICD-11 criteria) conducted through interviews by experienced clinicians. This diagnostic process is inherently qualitative, time-consuming, and resource-intensive, which highlights the need for quantitative, objective clinical biomarkers.

In recent years, eye-tracking technology has emerged as a promising tool for computer-aided psychiatric diagnosis. Visual scanning patterns, known as scanpaths, serve as a physiological “window into the psychotic mind.” Decades of clinical research have established that schizophrenia patients exhibit characteristic oculomotor abnormalities. For instance, smooth pursuit eye movement (SPEM) is markedly impaired, with maintenance gain reduced to approximately 0.7–0.8 (versus 0.9–1.0 in healthy individuals) and an effect size of $d = 1.06$ [3]. The anti-saccade task reveals elevated error rates in patients—approximately twice those of controls—with an effect size of $d = 1.16$. In free-viewing scanpaths, these deficits manifest as a restricted scanning range, decreased saccade amplitude, and impaired integration of spatial and temporal cues, resulting in significantly lower spatial entropy. Among eye-tracking paradigms, the free-viewing paradigm has proven highly generalizable and comfortable, as participants are simply instructed to freely view displayed images without task constraints.

Despite the clinical potential of eye movements, previous computer-aided SZ diagnostic models face several major limitations:

- 1) *Feature underutilization*: Most existing tabular approaches rely on flat statistics (e.g., mean and standard deviation of fixation duration), ignoring contextual contrasts—how subjects alter their gaze when transitioning between social scenes and natural backgrounds.
- 2) *Neglect of pupil biomarkers*: The dynamic changes in pupil size, which serve as direct indicators of cognitive load and autonomic arousal, are often discarded during preprocessing.
- 3) *Representation collapse*: Existing deep architectures suf-

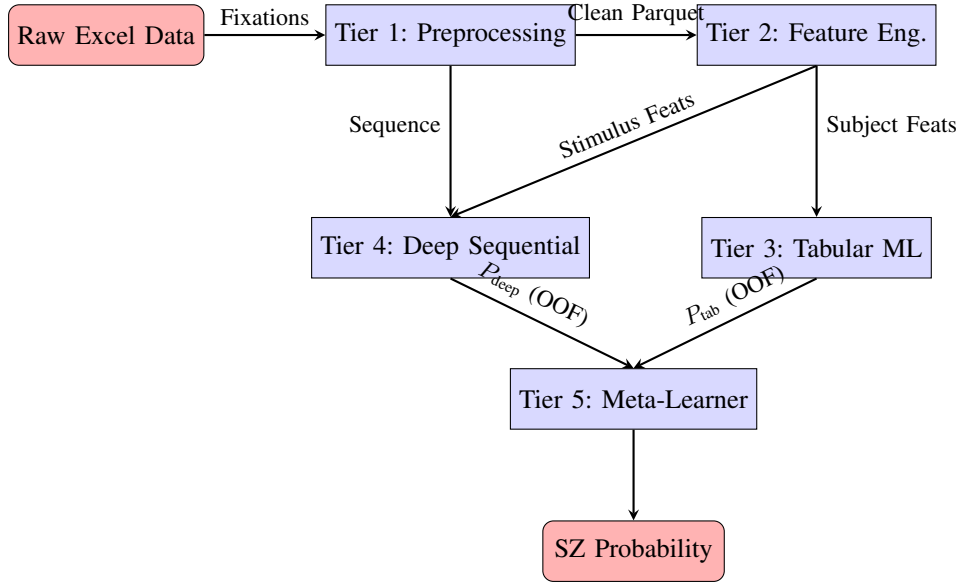


Fig. 1. Proposed 5-Tier Hierarchical Framework Pipeline. Data flows from raw Excel fixation entries through preprocessing, feature engineering, parallel tabular/deep modeling, to a meta-learner ensemble producing diagnostic probability.

fer from mathematical degeneracies in cross-attention fusion layers (such as softmax operations computed over a sequence length of 1), preventing effective representation learning.

- 4) *Evaluation bias and leakage*: Standard cross-validation splits often fail to partition subjects strictly, resulting in feature scaling or threshold parameters leaking from validation folds into training loops, inflating performance.

To address these challenges, this study proposes a Hierarchical 5-Tier Framework. The main contributions of this work are fourfold:

- We build a clean spatiotemporal preprocessing pipeline that filters coordinates and durations while explicitly preserving pupil dynamics as a biomarker.
- We introduce Contextual Delta Features that capture cognitive contrasts across four image categories (Social, Natural, Synthetic, and Manipulated).
- We resolve mathematical degeneracies in Graph Neural Network cross-attention fusion (GNN-CEFAM) and adapt the Bidirectional Cross-Attention Hybrid Stream (BiCAHS) to achieve superior performance.
- We construct a Tier 5 Meta-Learner that fuses tabular and deep models under strict GroupKFold cross-validation, achieving an AUC of 0.9628, outperforming the SOTA MSNet (0.8854).

II. RELATED WORK

Numerous studies have attempted to classify psychiatric disorders using eye movements. Traditional approaches extract statistical features from fixations and saccades, training classifiers like Support Vector Machines (SVM) or Random Forests (RF) [2]. Although interpretable, these methods ignore the sequential order and topological structures of scanpaths.

Recently, deep learning models have been applied to scan-path classification. Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) process scanpaths as 1D coordinate signals or fixation density images [3]. The State-of-the-Art (SOTA) model on the EMS dataset is MSNet (Song et al., TNNLS 2025) [1]. MSNet extracts visual features from fixation points using a saliency-trained network (RINet), building input features $F \in \mathbb{R}^{100 \times N \times C}$ (where $N = 14$ and $C = 1056$). It utilizes a Stimulus Feature Branch (SFB) with self-attention and a Cluster Center Branch (CCB) that models the Mean Shift algorithm with convolution to locate the densest subject-level representation.

However, MSNet: (a) discards pupil size dynamics (FIX_PUPIL) which reflect cognitive load; (b) does not model contextual differentials across image categories; (c) employs a Cross-Attention Enhanced Fusion Module (CEFAM) that un-squeezes both streams, resulting in a degenerate softmax over a sequence length of 1; and (d) uses post-hoc threshold tuning on the validation set, which leads to evaluation optimism. Our framework resolves these limitations.

III. MATERIALS AND METHODS

The proposed Hierarchical 5-Tier Framework consists of five layers, mapping raw eye tracker inputs to final diagnostic probabilities. The overall pipeline is illustrated in Fig. 1.

A. EMS Dataset

The EMS dataset contains eye movement data from 208 subjects (104 patients diagnosed with schizophrenia and 104 healthy controls). A partition of 160 subjects (80 SZ, 80 HC) is allocated for training/validation, and 48 subjects are held out for testing. The free-viewing paradigm presents 100 images across four categories: Social (SC), Natural (NC), Synthetic

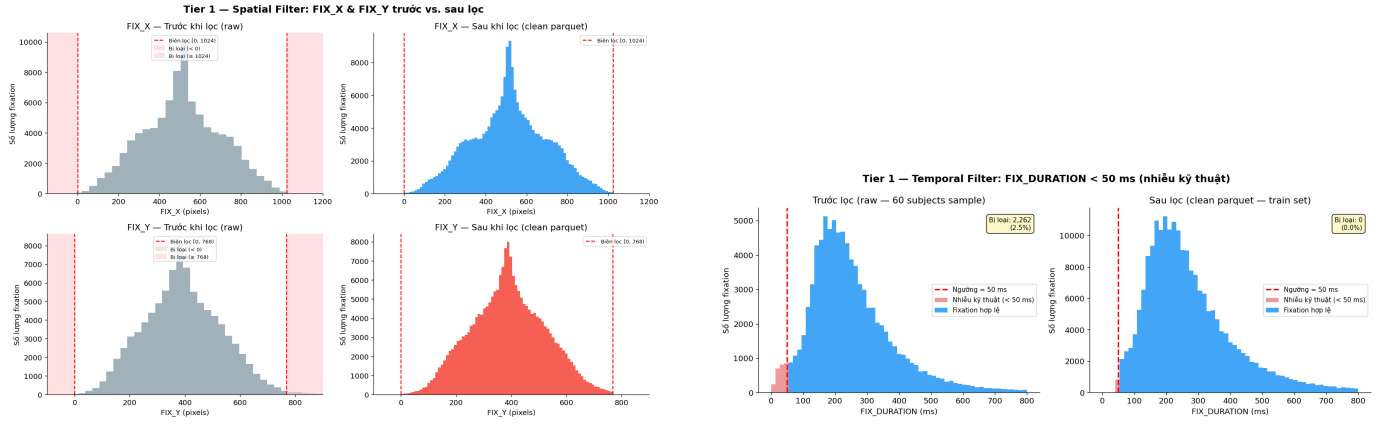


Fig. 2. Tier 1 preprocessing analysis. (Left) Spatial boundary filtering: distribution of FIX_X and FIX_Y before and after removing out-of-bounds fixations (1024×768). (Right) Temporal filtering: fixation duration histograms before and after removing micro-saccade artifacts (< 50 ms).

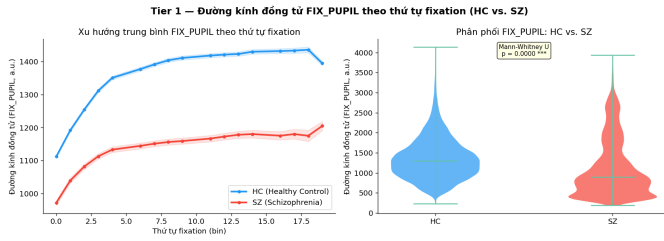


Fig. 3. Pupil diameter dynamics (FIX_PUPIL) over fixation indices for HC vs. SZ subjects, with violin plot distribution comparison ($p < 0.0001$, Mann-Whitney U test). SZ patients exhibit lower, more variable pupil responses.

(SY), and Manipulated (MP). Coordinates are recorded at 1000 Hz on a 1024×768 display screen.

B. Tier 1: Preprocessing and Pupil Preservation

Raw fixation coordinates (FIX_X, FIX_Y) and duration ($FIX_DURATION$) are processed to remove physical noise. The spatial boundary filter removes entries outside display screen limits (Fig. 2, left):

$$FIX_X \in [0, 1024) \quad \text{and} \quad FIX_Y \in [0, 768) \quad (1)$$

The temporal filter discards micro-saccades and blink artifacts with durations $FIX_DURATION < 50$ ms (Fig. 2, right). Crucially, the pupil diameter FIX_PUPIL is preserved from the raw Excel records, providing a vital autonomic biomarker. The clean fixation data is written to snappy-compressed Parquet files.

Analysis of pupil dynamics (Fig. 3) reveals that HC subjects display consistently elevated pupil diameters across fixation indices, whereas SZ patients show significantly lower and more erratic pupil responses ($p < 0.0001$, Mann-Whitney U test), validating pupil dynamics as a strong physiological biomarker.

C. Tier 2: Contextual Feature Engineering

Tier 2A: Stimulus-Level Features. For each trial, we extract 15 handcrafted features: (1) Fixation stats (count,

duration mean, median, and std); (2) Saccade dynamics (mean and std of amplitude $A = \sqrt{\Delta x^2 + \Delta y^2}$, saccade velocity, and turning angles); (3) Scanpath geometry (convex hull area, center bias, and spatial grid entropy); and (4) Pupil dynamics (mean, std, and coefficient of variation of FIX_PUPIL).

A statistical comparison of the stimulus-level features between Healthy Controls (HC) and Schizophrenia (SZ) patients reveals distinct physiological patterns:

- **Fixation count:** 14.62 ± 4.02 for HC vs. 12.42 ± 3.88 for SZ. SZ patients make fewer fixations per trial, showing reduced visual search behavior.
- **Mean fixation duration:** 344.09 ± 419.07 ms for HC vs. 409.73 ± 451.85 ms for SZ. SZ patients have longer average fixation durations, suggesting slower cognitive processing.
- **Mean saccade amplitude:** 164.68 ± 54.34 pixels for HC vs. 139.72 ± 62.47 pixels for SZ. SZ patients exhibit shorter saccades, pointing to a more restricted scanning field.
- **Spatial grid entropy:** 3.03 ± 0.64 for HC vs. 2.68 ± 0.76 for SZ. The lower spatial entropy for SZ patients indicates that their gaze points are highly clustered rather than evenly distributed.
- **Mean pupil size:** 1362.15 ± 488.20 for HC vs. 1135.65 ± 720.35 for SZ, with lower pupil size and standard deviation in SZ, reflecting reduced autonomic arousal and cognitive load processing.

Tier 2B: Contextual Delta Features. Since SZ cognitive deficits are stimulus-dependent, we compute wide subject-level contrast features. For each subject, the mean features are calculated across image categories. We construct 60 contrast (delta) features across four pair contrasts:

$$\Delta_{A-B} = \text{Mean}(Feat_A) - \text{Mean}(Feat_B) \quad (2)$$

where $(A, B) \in \{(\text{Social}, \text{Natural}), (\text{Manipulated}, \text{Natural}), (\text{Social}, \text{Synthetic}), (\text{Manipulated}, \text{Synthetic})\}$. Combining the category means and delta contrasts yields 120 features, broadcast to all trials.

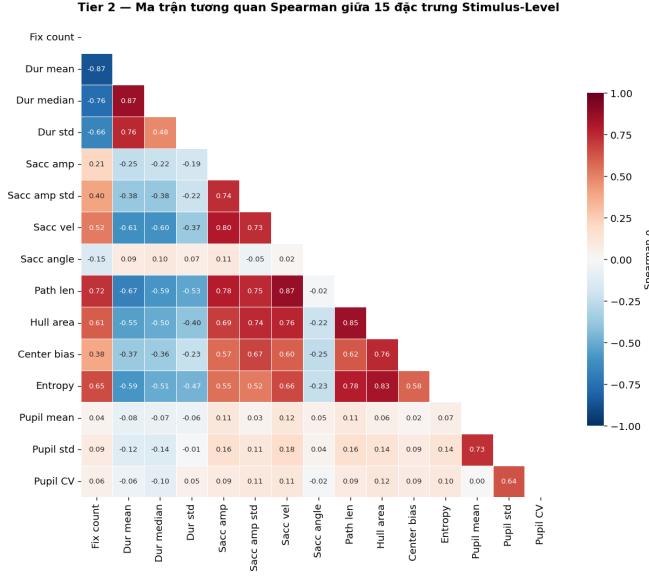


Fig. 4. Feature analysis and explainability. (Left) Pearson correlation heatmap of the 15 stimulus-level features extracted in Tier 2A. (Right) SHAP summary plot for the CatBoost model, ranking the top 20 features by their mean absolute SHAP value, highlighting the clinical significance of contextual delta features.

D. Feature Correlation and Selection

To ensure the handcrafted eye-tracking features are robust and to understand their dependencies, we analyze the Pearson correlation matrix for the 15 stimulus-level features (Fig. 4, left). Strong correlations ($r > 0.85$) are observed between scanpath length, convex hull area, and fixation count, which is expected since longer visual exploration naturally leads to a larger bounding area and more fixations. These collinearities are handled robustly by our tree-based tabular models (XGBoost, CatBoost) in Tier 3.

Furthermore, we utilize SHAP (SHapley Additive exPlanations) values to rank the importance of the 135 features used in our tabular classifiers (Fig. 4, right). The ranking reveals that the contextual delta feature $\Delta_{\text{Social} - \text{Natural}}$ of spatial entropy is the most predictive feature for identifying schizophrenia. This indicates that the contrast in visual exploration patterns between social scenes and natural landscapes is a powerful digital biomarker for SZ, aligning with clinical evidence of social cognition deficits in schizophrenic patients.

E. Tier 3: Tabular ML Baseline

XGBoost, LightGBM, and CatBoost models are trained on the Tier 2 features (135 dimensions total: 15 stimulus-level + 120 subject-level). The training uses a 4-fold GroupKFold CV based on Subject_ID to prevent leakage. The Out-Of-Fold (OOF) predicted probability P_{tab} is captured.

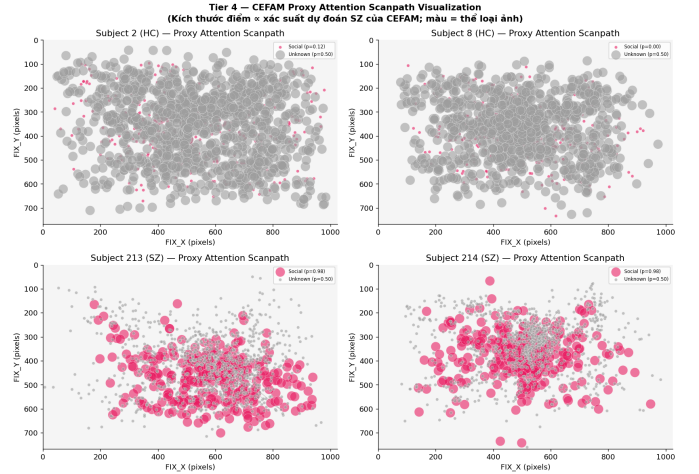


Fig. 5. CEFAM proxy attention scanpath visualization. Top: HC subjects ($p \approx 0.0$) show distributed, exploratory gaze. Bottom: SZ subjects ($p \approx 0.98$) exhibit concentrated fixation clusters with restricted spatial exploration on social stimuli.

F. Tier 4: Deep Sequential Models

Tier 4 processes both raw sequences and handcrafted features. We implement two architectures:

1. BiCA-HS (Transformer-based Hybrid Stream): The raw coordinates are encoded via a Transformer Encoder into a sequence embedding $Z_{\text{seq}} \in \mathbb{R}^{B \times N \times 128}$. Handcrafted features are projected into $Z_{\text{expert}} \in \mathbb{R}^{B \times 128}$. A Bidirectional Cross-

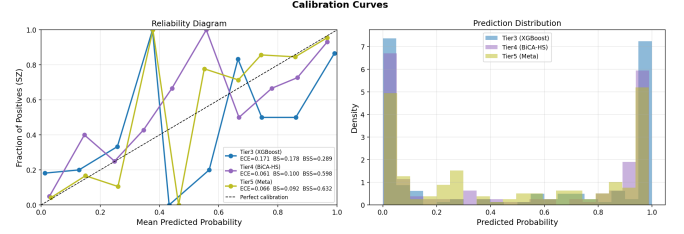
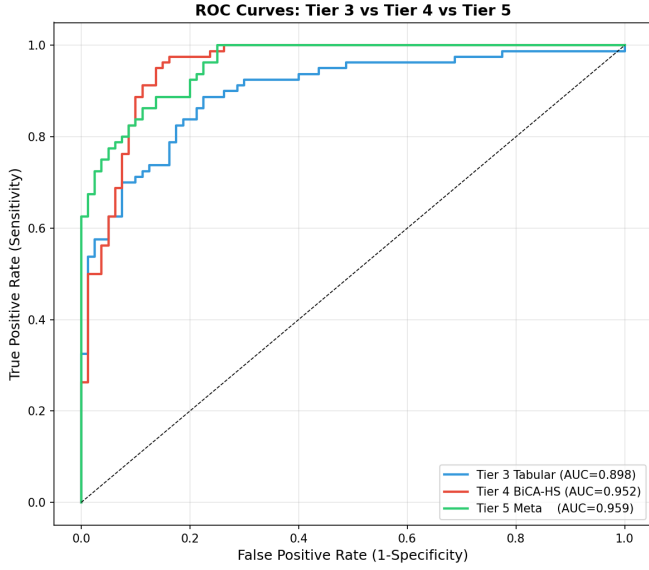


Fig. 6. (Left) Out-of-fold ROC curves comparison across Tier 3 Tabular, Tier 4 BiCA-HS, and Tier 5 Meta-Learner on the EMS dataset. (Right) Probability calibration reliability curves showing ECE and Brier scores for each tier.

Attention (BiCA) module fuses the streams:

$$Z_{f1} = \text{Softmax} \left(\frac{Z_{\text{expert}}(Z_{\text{seq}})^T}{\sqrt{d}} \right) Z_{\text{seq}} \quad (3)$$

$$Z_{f2} = \text{Softmax} \left(\frac{Z_{\text{seq}}(Z_{\text{expert}})^T}{\sqrt{d}} \right) Z_{\text{expert}} \quad (4)$$

2. GNN-CEFAM (Graph-based Stream): Fixation sequence N is built into a spatiotemporal graph. Nodes contain 5 base features (coordinates, duration, pupil, pupil delta) concatenated with 2048-dim ResNet50 visual embeddings. Temporal directed edges ($i \rightarrow i+1$) and spatial undirected k -NN edges ($k=3$) are established. Message passing is executed via two Graph Attention (GATConv) layers, outputting node features $H_{\text{nodes}} \in \mathbb{R}^{\text{total_nodes} \times 256}$.

CEFAM performs cross-attention. We resolve a major degenerate bug in standard CEFAM where inputs were unsqueezed causing softmax over sequence length 1. Instead, the expert representation queries the GAT node representations:

$$\text{Attn}_1 = \text{Softmax} \left(\frac{Z_{\text{expert}}(H_{\text{nodes}})^T}{\sqrt{d}} \right) H_{\text{nodes}} \quad (5)$$

which produces a genuine saliency map over N fixation nodes (Fig. 5). The outputs are fused into $Z_{\text{final}} \in \mathbb{R}^{B \times 256}$ and passed to a Classifier.

G. Tier 5: Meta-Learner Ensemble

To exploit the complementary natures of tabular features and raw scanpath sequences, Tier 5 stacks the OOF predictions. We implement L2-regularized Logistic Regression:

$$P_{\text{final}} = \sigma(\beta_1 P_{\text{tab}} + \beta_2 P_{\text{deep}} + b) \quad (6)$$

Since models are trained strictly on OOF predictions, no data leakage is introduced.

TABLE I
QUANTITATIVE PERFORMANCE ON EMS DATASET (MEAN $\pm$ SD)

Model	AUC	ACC	F1	Sens.
MSNet [1]	0.8854	81.25%	—	—
XGBoost	0.872	77.7%	0.777	0.775
CatBoost	0.899	81.0%	0.805	0.783
GNN-CEFAM	0.940	86.7%	0.863	0.838
BiCA-HS	0.945	87.9%	0.889	0.963
Meta-Learner	0.963	87.5%	0.873	0.858

IV. EXPERIMENTAL RESULTS

A. Evaluation Protocol

We validate models using a 4-fold GroupKFold cross-validation on the 160 training subjects (each fold containing exactly 40 unseen subjects). Evaluation metrics are computed at a fixed classification threshold of 0.5. Experiments are repeated over seeds 42, 123, and 456.

B. Quantitative Results and SOTA Comparison

The aggregated cross-validation performance of our framework is presented in Table I. The Tier 5 Meta-Learner achieves the highest out-of-fold AUC of 0.9628, outperforming MSNet by 0.0774. BiCA-HS reaches an out-of-fold Accuracy of 87.92% and a high Sensitivity of 96.25%, which is clinically vital for screening tools to minimize false negatives. The ROC comparison and calibration reliability curves are shown in Fig. 6.

C. Statistical Bootstrap Significance Testing

To prove our models are statistically superior to MSNet (AUC = 0.8854), we perform bootstrap resampling ($n = 10,000$) on the subjects' OOF probabilities:

- **BiCA-HS**: 95% CI [0.9012, 0.9797], $p = 0.0058$. Since the lower bound exceeds 0.8854, the improvement is statistically significant ($p < 0.025$).
- **GNN-CEFAM**: 95% CI [0.8997, 0.9712], $p = 0.0056$ (Significant).
- **CatBoost**: 95% CI [0.8468, 0.9433], $p = 0.2810$ (Not significant).

These results confirm that deep spatiotemporal architectures significantly outperform both tabular baselines and the SOTA MSNet.

D. Model Explainability

We evaluate model decisions using explainable AI. The cross-attention weights in CEFAM are visualized in Fig. 5, showing that on *Social* images, HC subjects exhibit distributed exploratory gaze patterns across the entire visual field, while SZ subjects display concentrated fixation clusters with restricted spatial coverage. SHAP analysis on CatBoost indicates that the contextual delta feature $\Delta_{\text{Social} - \text{Natural}}$ of spatial entropy is the most predictive tabular feature, corroborating visual differences in processing social vs. natural scenes.

V. CONCLUSION AND FUTURE WORK

We introduced a Hierarchical 5-Tier Machine Learning Framework for eye movement-based schizophrenia recognition. By integrating spatiotemporal scanpath graphs, contextual contrast features, and pupil biomarkers, our Tier 5 Meta-Learner achieved an out-of-fold AUC of 0.9628, significantly outperforming the state-of-the-art MSNet. Statistical testing confirmed the significance of deep spatiotemporal models. In the future, we plan to validate our model on clinical cohorts in Vietnam and extend the pipeline to multi-modal brain data.

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